

In emergencies such as fires or gas leaks, immediate and organized evacuation is crucial. Trained personnel, known as responders, are deployed to sweep the affected building and rescue occupants who may still be trapped. The presence of time-varying hazards necessitates the implementation of effective sweeping strategies to ensure the safety of both occupants and responders, striving to leave no one behind.

The problem emphasizes that building layout and the limited information available to responders are critical factors in designing sweeping strategies. This paper employs mathematical modeling to develop a general search strategy capable of adapting to diverse building environments.

To address the first requirement, we analyze the scope and key elements of the model. Critical factors—including room clearance criteria, responder deployment, building sweep priority, redundancy principles, and emergency constraints—are clearly defined.

Next, a Localized Model was developed to simulate responder behavior within individual rooms. By comparing different search approaches, it was found that combining complementary strategies—nearest-first and farthest-first—enhanced both search efficiency and completeness.

At the macroscopic level, a Global Building Clearance Model was constructed to abstract entire building layouts. It dynamically optimizes rescue routes by integrating fire spread parameters, including both fire and smoke propagation. A comprehensive fire-smoke-toxicity analytical model was incorporated to quantify elapsed time during the rescue operation.

For the third requirement, multi-scenario analyses were conducted to validate model applicability, covering single-floor, multi-room, and multi-floor buildings. Path planning strategies were compared to identify optimal solutions under different scenarios. Best Team Optimization analyses determined the optimal number of responder teams and identified the most effective strategies for situations with limited personnel. Notably, the allocation of responders had a greater impact on rescue outcomes than simply increasing team numbers.

Further data collection and calibration were performed, drawing on authoritative databases for critical parameters such as fire growth coefficients and smoke characteristics. This enhanced the realism of the simulations and validated the applicability of the models across multiple scenarios.

An Agent-Based Simulation Model was developed to visually demonstrate the rescue process under various conditions, including time-varying hazards. This model confirmed that a two-stage search strategy—comprising a preliminary rapid inspection followed by secondary detailed confirmation—achieves an optimal balance between efficiency and accuracy, providing scientific guidance for real-world firefighting and rescue operations.

Finally, sensitivity analysis was conducted to assess model stability. The results revealed that key parameters, particularly smoke duration, significantly influence rescue outcomes, confirming that the model remains robust under realistic and variable conditions.

Table of Contents

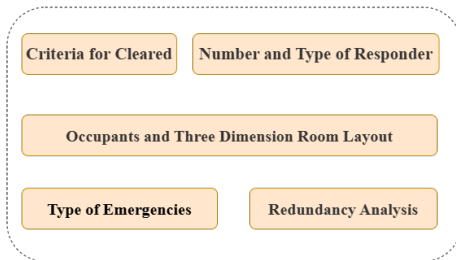
1	Introduction	3
1.1	Problem Restatement.....	3
1.2	Assumptions and Justifications	4
1.3	Table of Variables	4
2	Scope and Key elements of the model.....	5
2.1	Criteria for Cleared.....	5
2.2	Personnel, Layout, and Redundancy Principles	6
2.3	Number and Type of Responder.....	8
2.4	Type of Emergency.....	9
3	Model.....	9
3.1	Localized Responder Sweep Model	10
3.1.1	Modeling Responder Behavior in Single Room	10
3.1.2	Comparison of Sweeping Strategies	11
3.2	Global Building Clearance Model.....	12
3.2.1	Based Model towards A Simple Room Layout.....	12
3.2.2	Consideration of the α Parameter.....	14
3.2.3	Multi-scenario Analysis	16
3.2.4	Team Optimization.....	17
3.2.5	Reduced Team Approach	18
4	Data Analysis and Model Extension	18
4.1	Multi-Source Data Collection and Feature Analysis	18
4.2	Agent-Based Simulation Model	19
4.2.1	One-Floor Simulation	20
4.2.2	Two-Floor Simulation	20
5	Sensitivity Analysis.....	21
5.1	Localized Responder Sweep Model Analysis	21
5.2	Agent-Based Model Analysis	22
6	Conclusion.....	23
6.1	Strengths and Weaknesses	23
6.2	New Technologies to Consider.....	24
	Reference.....	24
	Letter to the Local Emergency Planning Committee	25
	AI Report	26

1 Introduction

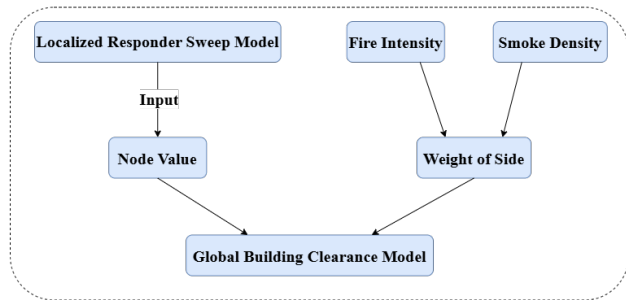
Emergencies such as fires or gas leaks require immediate and organized evacuation. Trained personnel, often called **responders**, are deployed to sweep the affected building, checking for **occupants** who may still be trapped and rescuing them. Given the presence of time-varying hazards, effective sweeping strategies must be implemented to ensure the safety of both occupants and responders.

1.1 Problem Restatement

Part 1: Definition of Scope and Key elements



Part 2: Building models and Find the Optimal Results



Part 3: Data Analysis and Model Extension

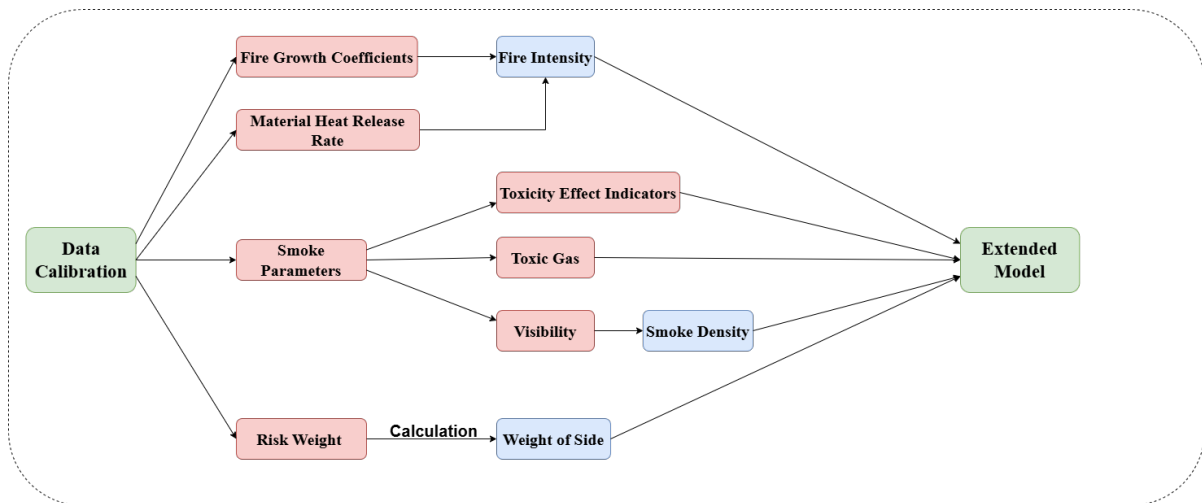


Fig. 1: Workflow of Our Work

The goal of this project is to develop multiple models to optimize workflow and strategies under different fire and rescue scenarios. Firstly, we are asked to define standard criteria for a single responder, such as how to determine whether a room has been fully cleared and the specific tasks assigned to each responder. The environment also needs to be created, including the overall building layout as well as the layouts of individual rooms, such as children's daycare centers, office spaces, and warehouses. Optimal sweeping strategies are then identified through analysis and comparison, and the overall sweeping time is calculated based on these strategies.

Further, additional factors are incorporated to better simulate real-world conditions. These

include time-varying hazards such as the spread of fire and smoke, a limited number of responder teams, and varying priorities for different room types. We also apply the model to new building layouts with visualizations to illustrate the responders' routes during the rescue operations. We also consider new technologies to assist the responders in the sweep.

1.2 Assumptions and Justifications

Assumption 1: The emergency (e.g. fire) does not damage the structure of the building.

Justification 1: We consider the layout of the building as static, ignoring the factor of a probable collapse or other irreversible changes, which may multiply the difficulty of the sweep operation.

Assumption 2: An occupant found by responders is noted as 'Rescued'.

Justification 2: We assume the occupants can't be harmed by the remaining fire when escaping. With the help of safety equipment provided by responders, occupants can all be rescued safely after being found.

Assumption 3: Responders don't put out the fire when sweeping the building.

Justification 3: In most fire incidents, fatalities are more often caused by smoke inhalation than by the flames themselves, as occupants typically escape from the fire and take refuge in a room. Therefore, the primary objective of responders is to locate and rescue the occupants. Extinguishing the fire can delay this process, and smoke may continue to pose a hazard even after the flames are put out.

Assumption 4: Responders prioritize their own safety when they recognize that fire or smoke conditions pose a life-threatening risk.[1]

Justification 4: Since responders are equipped with professional protective gear, they are expected to withstand conditions that would be lethal to ordinary occupants. However, if the fire or smoke becomes too severe for the responders to advance safely, they may be unable to proceed, potentially resulting in no occupants being rescued.

1.3 Table of Variables

Symbol	Definition
I	Fire Intensity
ρ	Smoke Density
R	Smoke Radius
r	Spread Rate
k_ρ	Attenuation Coefficient of smoke density
d	Distance between fire source and current position
W	Weight of an edge connecting two nodes

C_D	Influence of Smoke Density on the edge weight
C_I	Influence of Fire Intensity on the edge weight
α	Fire Watt Growth Constant
$\dot{Q}$	Heat Release Rate
t	Time Elapsed since the ignition of the fire
Y_s	Mass of Smoke generated per unit of energy released
k_{map}	Flame Spread Constant
E	Congestion Efficiency
$C_{crowding}$	Crowding Coefficient
$N_{responder}$	Number of Responders in the same hallway

2 Scope and Key elements of the model

2.1 Criteria for Cleared

This defines the status of the room whether it is swept by the fire fighters. We assumed that sweep could find all the occupants. Due to high dependency on available time and clearance strategies, we divide the sweep as two stages: **Primary and Secondary Sweep**.

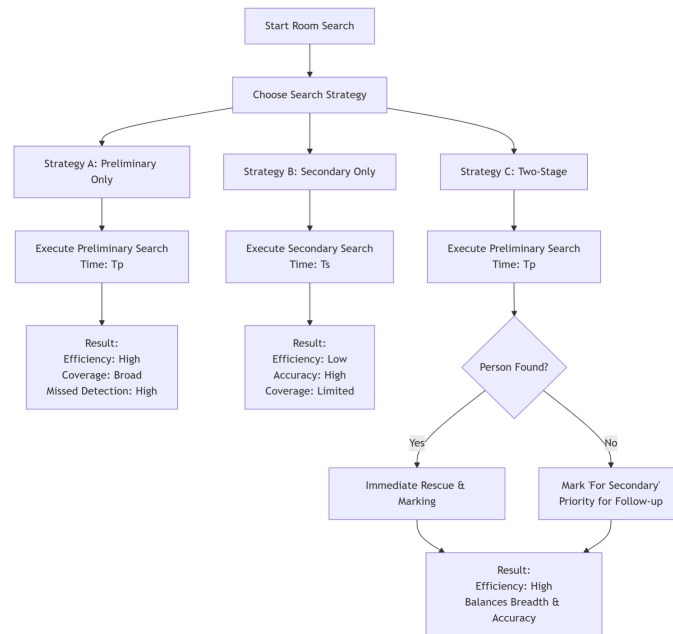


Fig. 2: Three Strategic Options for Room Search and Their Implementation Procedures

1. Primary Sweep -- Quick Confirmation

Responders perform a quick but systematic visual inspection from the doorway or by entering the room, checking for trapped individuals or responses to calls. Suitable for low-risk or simply laid-out rooms. Besides, the responders have to shout loudly (e.g. "Is anyone there? Fire Department!") and listen for a response. This is very important in low-visibility conditions.

2. Secondary Sweep -- Detailed Confirmation

Responders have to check behind obstacles, under desks, inside cubicles, restroom stalls, and all areas where a person might hide or be unconscious, which requires more time. Once one room is cleared, responder would use a standardized mark (e.g. fluorescent X-shaped sticker or chalk mark) on the door to prevent re-checking and inform other responders.

The reason for adopting the "preliminary-secondary search" two-stage strategy is to quickly screen high-value targets through initial investigations, thereby optimizing resources and maximizing rescue efficiency.

3. Strategy Comparison

The diagram below intuitively demonstrates the differences and advantages of the three strategies through the decision-making process for a single room: Mathematical Efficiency Comparison. The key criterion for decision-making is the rescue efficiency per unit of time, calculated as follows:

$$\text{Strategy A (Preliminary Only): Efficiency} = \frac{P_1 \cdot E_p}{T_p \cdot \text{Strategy}_B}$$

$$\text{Strategy B (Secondary Only): Efficiency} = \frac{P_1 \cdot E_s}{T_s \cdot \text{Strategy}_C}$$

$$\text{Strategy C (Two-Stage): Efficiency} = \frac{P_1[E_p + (1 - E_p) \cdot E_s]}{T_p + (1 - E_p) \cdot P_1 \cdot T_s}$$

Substituting specific values (assuming $P_1=0.2$, $T_p=10s$, $T_s=60s$, $E_p=0.7$, $E_s=0.98$), the conclusion becomes clear:

1. Strategy A (Preliminary Only) has moderate efficiency (0.014): While coverage is fast, the high rate of missed inspections results in a low overall rescue success rate.
2. Strategy B (Secondary Only) has the lowest efficiency (0.0033): Although the inspection is accurate, the overall rescue coverage is too limited due to the excessive time required.
3. Strategy C (Two-Stage) has the highest efficiency (0.0146): It intelligently allocates time to the areas where it is most needed, achieving the optimal balance between speed and thoroughness.

Therefore, the two-stage model ensures the highest overall rescue success rate for the system within a limited time.

2.2 Personnel, Layout, and Redundancy Principles

Room type, layout, and corresponding redundancy strategies are critical factors in determining search priority, estimating search time, and ensuring operational reliability. Different room types vary significantly in occupant self-rescue capability, spatial complexity, and risk level,

directly influencing the allocation of rescue resources, strategy selection, and safety assurance.

Fire Rescue - 3D Room Complexity Analysis

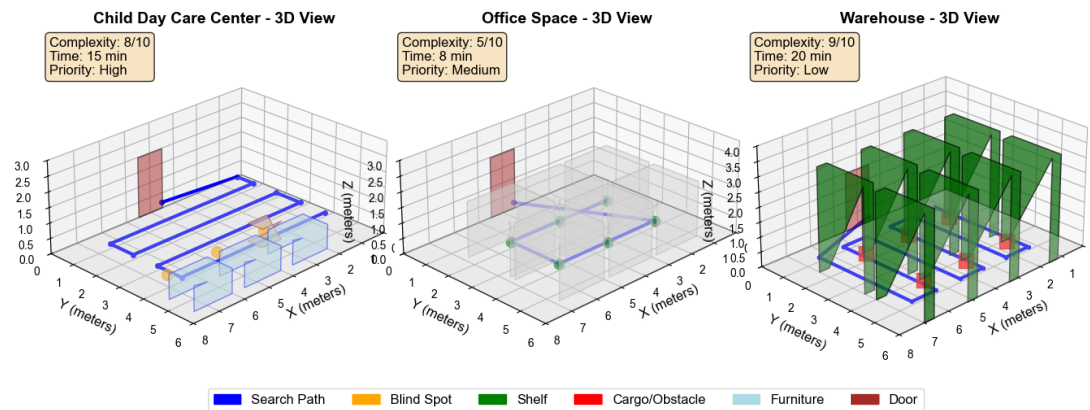


Fig. 3: Indoor environments with their respective search routes

Table 1 details the characteristics of room types and their impact on search operations:

Table 1: Characteristics of Different Room Types

Room Type	Occupant	Priority	Layout Features	Sweeping Assessment
Child Daycare Center	Children require assistance.	High	Multiple partitions; numerous hiding spaces under furniture.	Difficulty: High Time: Long Strategies: Thorough search of all hiding spots, time for reassurance, and implementation of spatial or verification redundancy .
Office Space	Mobile adults.	Medium-Low	Open-plan: Clear sightlines. Cubicle: Blocked sightlines.	Open Layout: Low difficulty/Short time Cubicle Layout: Medium difficulty/Medium time Strategies: Temporal redundancy in visually obstructed areas to ensure coverage.
Warehouse	Low density.	Low	Dense, high shelving and stacked goods; maze-like aisles with blind spots.	Difficulty: High Time: Long Strategies: Meticulous search of complex paths and blind spots, requiring spatial or verification redundancy for mitigation.

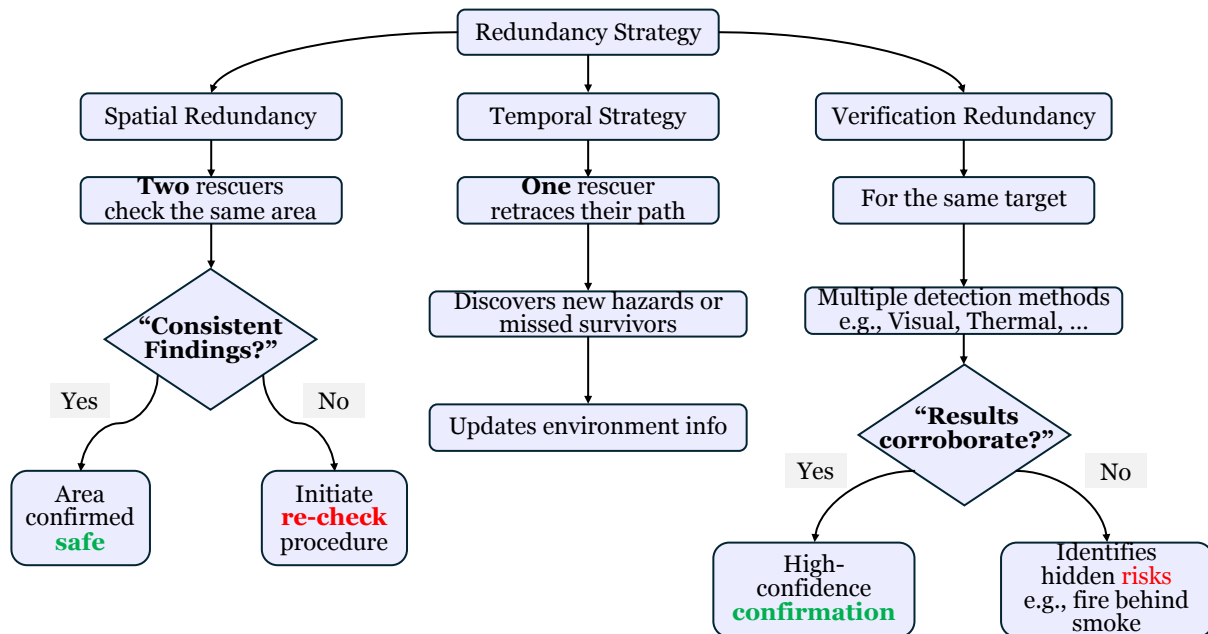


Fig. 4: Three Redundancy Strategies: Spatial, Temporal, and Verification

Explanation of Redundancy Principles[2]: Redundancy in post-fire search operations is a core design concept balancing safety and efficiency, primarily implemented in the following forms.

- 1. Spatial Redundancy:** Assigning two or more responders to inspect the same area, ensuring reliability through independent cross-verification.
- 2. Temporal Redundancy:** Requiring responders to retrace their original route at specific intervals or after task phases for secondary confirmation.
- 3. Verification Redundancy:** Combining multiple detection methods (e.g. visual inspection, thermal imaging, acoustic detection) in critical areas for result validation.

While redundancy strategies may impact operational efficiency to some extent, they are essential safety measures in high-risk areas (e.g. daycare centers, laboratories), low-visibility environments (dense smoke), and dynamic fire conditions.

2.3 Number and Type of Responder

The number and composition of responders are determined by factors such as building size, structural complexity, and the severity of the emergency. The model should be capable of calculating the optimal number of responders required for a given building layout to achieve efficient and safe operations.

Different specialization types play distinct roles in the rescue process. **Responders** serve as the primary responders, equipped with breathing apparatus and trained in search and rescue under smoke-filled conditions. **Hazardous materials teams** are essential in situations involving chemical spills or laboratory incidents, ensuring safe containment and mitigation of

risks. **Medical personnel** may enter either concurrently or immediately after the clearance phase to provide medical assistance to injured individuals. **Security personnel**, familiar with the building layout and daily occupancy patterns, can act as guides or take responsibility for specific zones, contributing to both coordination and efficiency in the overall rescue operation.

2.4 Type of Emergency

1. **Fire:** The main operational constraints include the spread of fire, the accumulation of smoke which progressively reduces visibility and safety, and the rise in temperature. Model has to incorporate a time-varying **hazard map** that updates as conditions evolve. Consequently, path planning must remain dynamic, continuously steering responders away from dangerous zones. Sweep operations must effectively compete with the pace of fire growth, making time a critical factor in determining whether a rescue can be completed safely and successfully.
2. **Gas leak:** the main constraints in this scenario involve **toxicity** and **explosion risk**, which require careful operational planning. Responders may need to shut off critical valves, making path selection dependent on access to these control points. While the overall risk distribution may appear more uniform than in fire scenario, entering high-concentration toxic areas demands specialized protective equipment. Additionally, **ventilation status** becomes a decisive factor in determining feasible routes.

3 Model

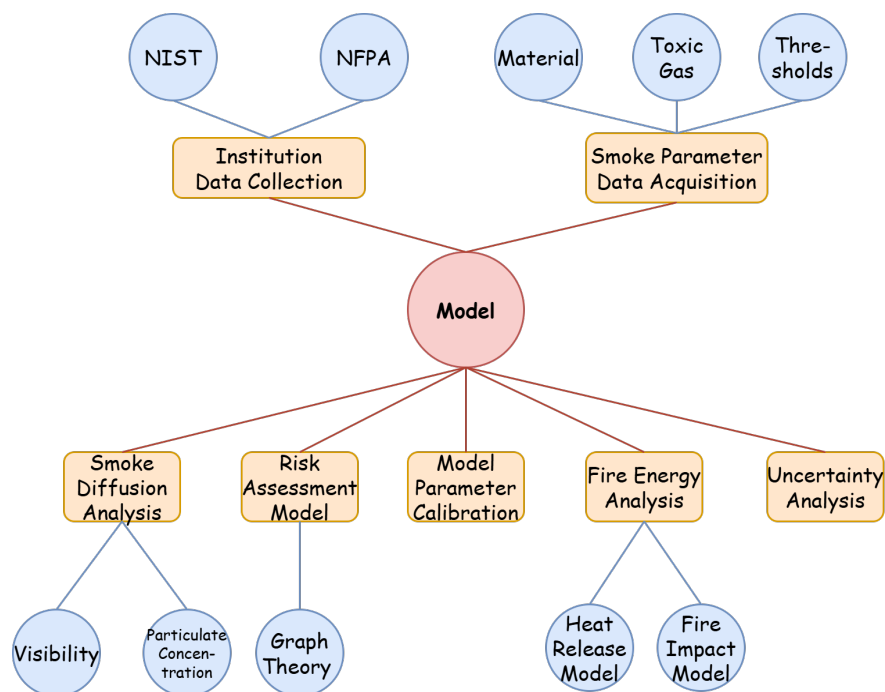


Fig. 5: Model Framework Composition

Before demonstrating our models, we have to state the relationship between models. The **Global Building Clearance Model** focuses on planning search strategies across the entire layout by abstracting the room layout into a graph. However, the actual search duration varies from room to room, which the global model alone cannot capture. Therefore, the **Localized Responder Sweep Model** simulates responders' actions inside a single room to generate realistic search times that serve as point values in the macro-model. Simultaneously, the micro-model allows us to refine and optimize room-level strategies, ensuring that the overall system reflects both large-scale planning and detailed operational behavior.

3.1 Localized Responder Sweep Model

3.1.1 Modeling Responder Behavior in Single Room

In this section, we address the internal dynamics of a **single-room search**. We develop a model to simulate the **room-level sweep process** and analyze the **decision-making behavior** of **responders** when locating occupants. This enables a more accurate estimation of the required operational time, which is later used as an input for the building-level model.

We construct a search model to simulate the actions of responders in the room of the type which mentioned in 2.2.

Two responders enter the room simultaneously but adopt different sweep strategies: one prioritizes distant areas (**Farthest First**), while the other focuses on nearby regions (**Nearest First**). As shown in **Fig. 6.1**, the blue responder moves directly toward the farthest parts of the room, while the red responder systematically clears the area around the entry point. Through this complementary division of labor, the team is able to sweep the entire room efficiently without overlooking any potential occupants.[3]

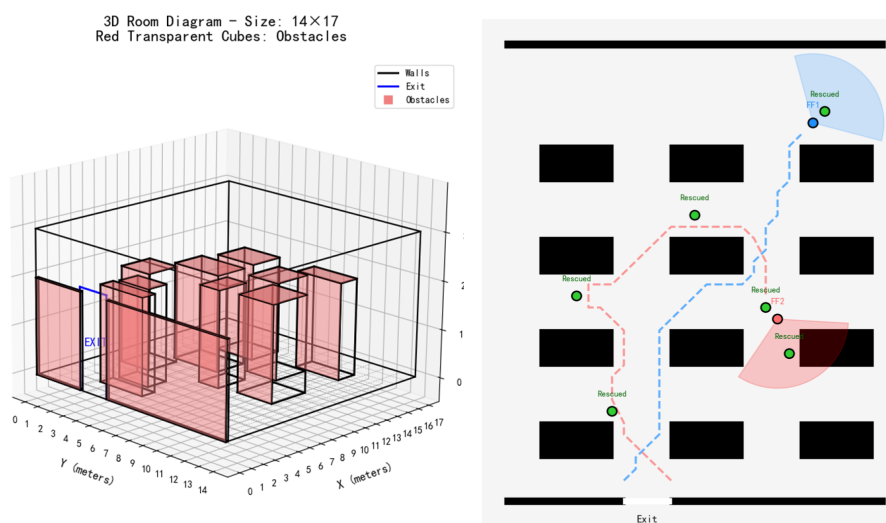


Fig. 6.1

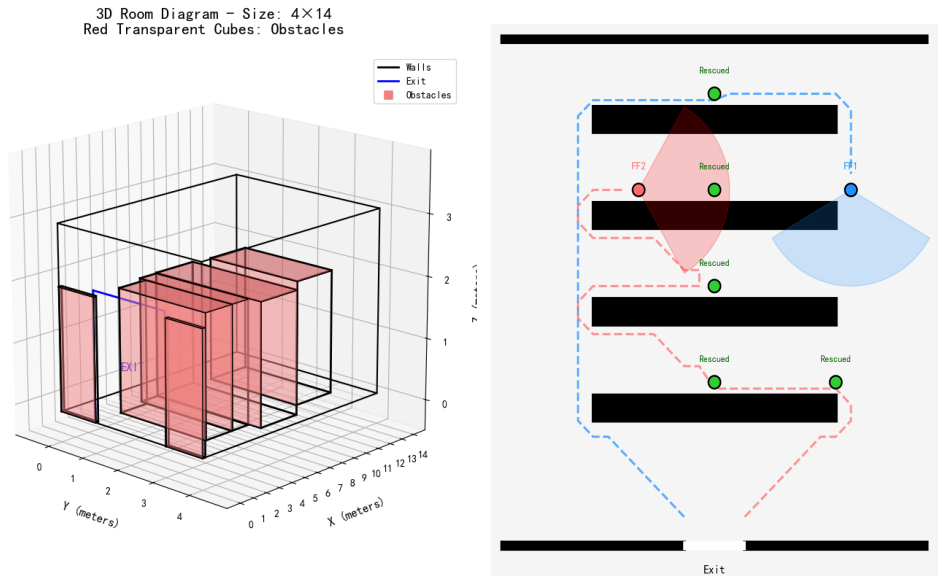


Fig. 6.2

Fig. 6: Simulation of a single room

In more complex indoor environments, larger obstacles can obstruct visibility and increase the likelihood of missing occupants. As shown in **Fig. 6.2**, the responder using the **Farthest First** strategy (blue) initially fails to detect an occupant positioned near a large obstacle. With support from the second responder (**Nearest First**) who sweeps the adjacent area the overlooked occupant is ultimately located and rescued. This scenario illustrates that combining complementary search strategies enhances both efficiency and completeness of the room sweep.

3.1.2 Comparison of Sweeping Strategies

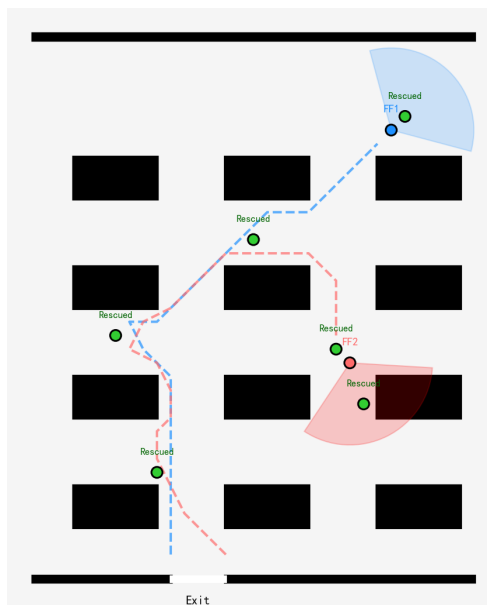


Fig. 7.1

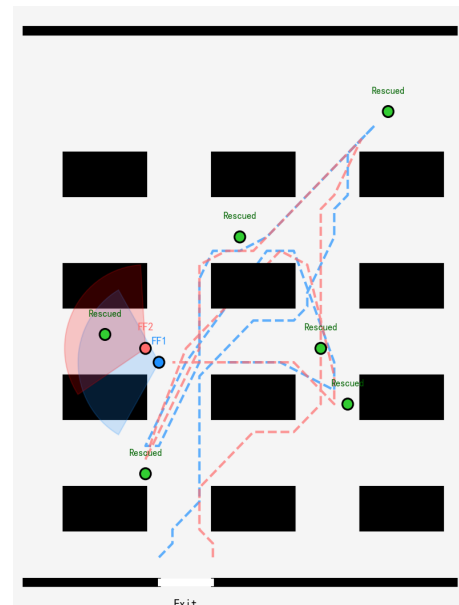


Fig. 7.2

Fig. 7: Simulation of responder routes under different search strategies

To evaluate the effectiveness of different sweeping strategies and combination, we analyzed scenarios in which two responders adopted **Nearest First** and **Spiral Search** patterns. Spiral Search prioritizes occupants located within the shortest radial distance, while Nearest First clears the closest accessible areas. However, the responders tend to converge on the same regions because both strategies share similar search principles. As shown in **Fig. 7.1**, this overlap causes their movement to resemble that of a single responder, reducing operational efficiency and increasing the risk of missing occupants in unvisited zones. Nevertheless, this pairing offers an advantage: responders remain close to each other, which can enhance safety by allowing immediate mutual support in the event of an unexpected hazard. Thus, the combination reveals a trade-off between **coverage efficiency** and **responder safety**.

Moving to the next situation, shown in **Fig. 7.2**, using the same strategy (**Farthest First**) results in more serious coordination issues. Possibly due to communication failure, both responders adopt identical search patterns. Although they initially appear to divide the room effectively—each sweeping one side—this coordination breaks down as the search progresses. When the responders converge toward the remaining occupants, their routes begin to overlap. One responder, even after efficiently clearing a table in the office, ends up covering the same zone as the other. This redundancy wastes valuable time and delays the clearing of unexplored areas, thereby increasing the risk for occupants still awaiting rescue. The scenario illustrates how identical strategies can undermine coordination in confined, obstacle-rich environments.

3.2 Global Building Clearance Model

3.2.1 Based Model towards A Simple Room Layout

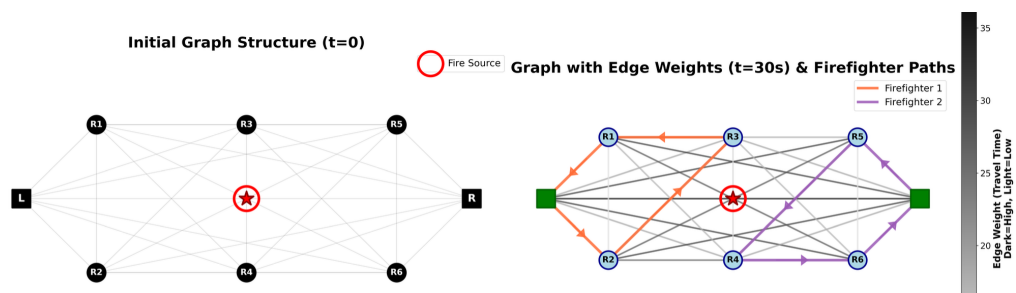


Fig. 8: Graph representation of base case

We constructed a model of a single floor with the fire source located at the center. As shown in **Fig. 8**, the floor layout is abstracted into a graph, where each node represents a room and each edge represents a pathway connecting two rooms. The weight of an edge reflects the difficulty or risk associated with traversing that hallway, with larger weights indicating greater danger. Each node is also assigned a value corresponding to the time required to search that specific room calculated from the previous chapter. In the figure, wider edges correspond to

higher weights. The colored path illustrates the route taken by the responder during the first stage, which involves a rapid sweep requiring only a short amount of time.

The weight of each edge is determined by the spread of fire and smoke, defined as follows:

$$I(pos, t) = I_0 * (1 - \frac{d}{R(t)})$$

$$\rho(pos, t) = I_0 * (1 - k_\rho * \frac{d}{R(t)}) \quad k_\rho = 0.7$$

$$R(t) = r * t$$

$$d = |pos - pos_{fire}|$$

where ρ represents the smoke density, R is the smoke radius (calculated as the product of time t and the spread rate r), k_ρ is the attenuation coefficient of smoke density and d denotes the distance between the fire source and the current position.

$$W(x_1, x_2, y_1, y_2, t) = W_{base} * (1 + \frac{D_{smoke}}{D_{max}} * C_D) * (1 + \frac{I_{fire}}{I_{max}} * C_I)$$

$$W_{base} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

The final weight W of an edge connecting two nodes is calculated as the product of the base weight and the influence factors of both smoke and fire. C_D and C_I is the coefficient representing the influence of smoke density and fire intensity on the edge weight. In our model, the weights of the edges vary over time. Both the fire and smoke simulations originate from a small central source and expand outward in a circular pattern. The intensity or influence of fire and smoke decreases linearly with increasing distance from the fire source. Unlike fire intensity, the rate of decrease in smoke density is smaller, indicating that the effective influence radius of smoke is larger than that of fire. This observation aligns with physical reality, as smoke generally spreads farther and affects a wider area than the flames themselves.

The next phase is the second-stage sweep, which takes more time and requires two responders to conduct a coordinated search. The visualization of the search has been shown in the chapter 3.1. To compare different search strategies, we aim to identify the most efficient approach. The **first method** employs a greedy algorithm, in which the responder always selects the edge with the lowest weight, representing a locally optimal search. The **second method** is based on a brute-force approach, which evaluates all possible paths to determine the shortest one. Alternatively, a **TSP-based strategy** (Traveling Salesman Problem) seeks the globally optimal route that minimizes the total travel cost while visiting all accessible rooms. In contrast, a slowest first method enables the longest sweep to be done and minimize the effect of spread of smoke and fire. Comparing these strategies allows us to evaluate the trade-off between computational efficiency and global optimality in the search process.

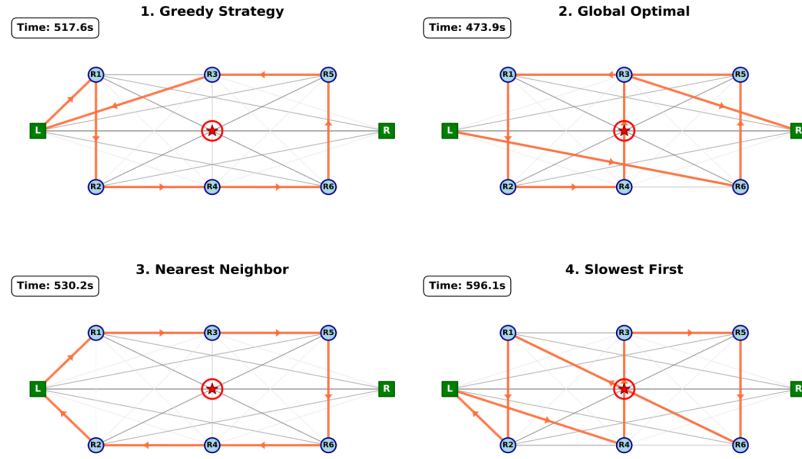


Fig. 9: Responder routes under different strategies

In the program, we use following table to have the simulation of the evacuation sweep with reference of data from National Institute of Standards and Technology (NIST) [4].

Table 2: Parameters used in model

	Fire intensity	Smoke density
Spread rate	0.0171 m/s	0.0257 m/s
Initial value	5.0	4.0
Weight coefficient	2.0	1.5

3.2.2 Consideration of the α Parameter

To make the model more realistic, we introduce a parameter α , referred to as the fire watt growth constant. A larger value of α indicates a faster rate of fire spread, as expressed by the following relationship.

$$\dot{Q}(t) = \alpha t^2$$

where Q denotes the heat release rate and t represents the time elapsed since the ignition of the fire. Both the fire intensity and smoke density are related to the heat release rate but follow different functional relationships, which can be described as follows:

$$\dot{m}_s = Y_s \cdot \dot{Q} \rightarrow \rho_{smoke} \propto \dot{m}_s^2$$

$$R(t) = k_{map} * \sqrt{Q(t)} \rightarrow I(pos, t) = I_0 * \left(1 - \frac{d}{k_{map} * \sqrt{Q(t)}}\right)$$

where $\dot{m}_s$ denotes the smoke generation rate, k_{map} represents flame spread constant and Y_s represents the mass of smoke generated per unit of energy released.

The following two graphs (**Fig 10** and **Fig 11**) illustrate the relationship between α and both smoke density and fire intensity, as well as their effects on the performance of different search strategies.

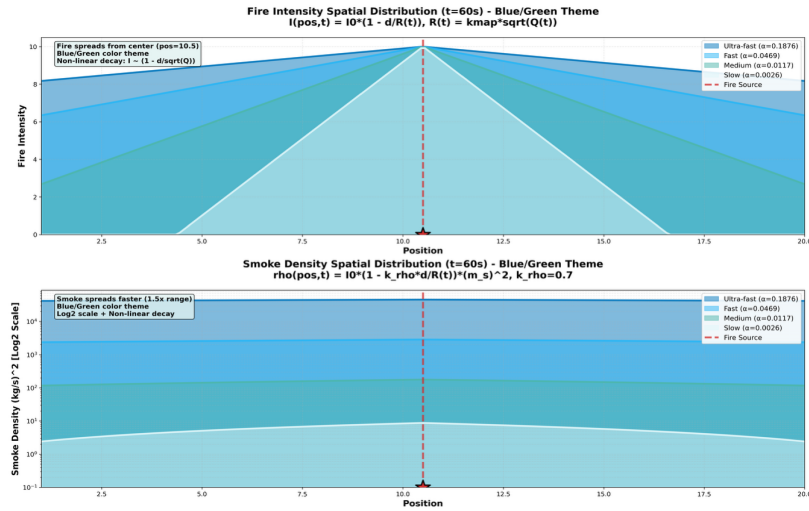


Fig. 10: Fire and smoke distribution under different α values

From the histogram below, the optimal strategy achieves the best overall performance, though it has the clear drawback of requiring a significantly longer computation time. The greedy and nearest-neighbor strategies demonstrate stable performance across different values of α , with gradual improvement and only minor deviations from the optimal solution. The slowest-first strategy appears to perform the worst in terms of total efficiency; however, it possesses an important potential advantage. By prioritizing the most time-consuming or high-risk areas at the beginning of the operation, this strategy allows responders to address the most dangerous zones before conditions deteriorate further, thereby improving safety and reducing the likelihood of entrapment in later stages of the mission.



Fig. 11: Time consumption of different strategies under varying α values

With further analysis, we compared the rescue time of a single firefighting team with the smoke spread time corresponding to the human tolerance threshold. Two responders were generally able to complete the rescue under gradually developing fires but were unable to do so in more severe scenarios. For intense fires with fast or ultra-fast α values, even three teams of responders could not complete the task in time. Although deploying more teams appears to improve efficiency theoretically, congestion within the building emerges as a critical factor that

significantly slows down the sweeping process. We will have an exploration further in later part. Nevertheless, three teams were able to successfully complete the rescue under fires with medium α values, which correspond to normal office fire scenarios.

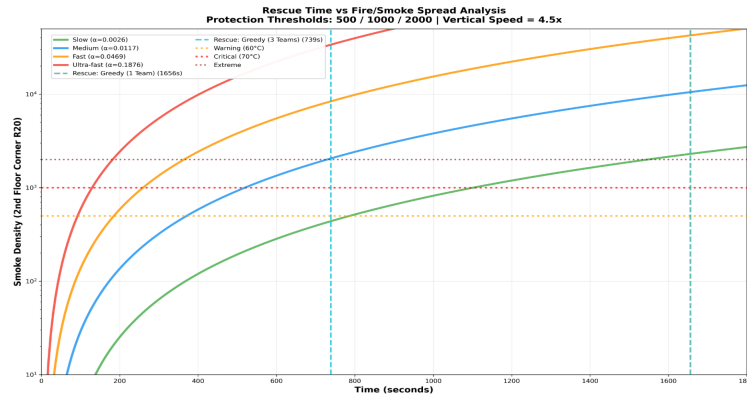


Fig. 12: Smoke density over time with warning and rescue thresholds

3.2.3 Multi-scenario Analysis

Now, we will develop different arrangement of rooms and apply our model to them. First, we still want to have a scenario for single floor. But in this time, we would have more rooms and alternative pathway for the fire fighter to sweep the floor shown in the **Fig 13**. This floor has 12 rooms in total with 4 columns and 3 rows and fire fighter could cross the room with extra time (product of penalty and point value of current room) instead of turning direction in the end of corridor.

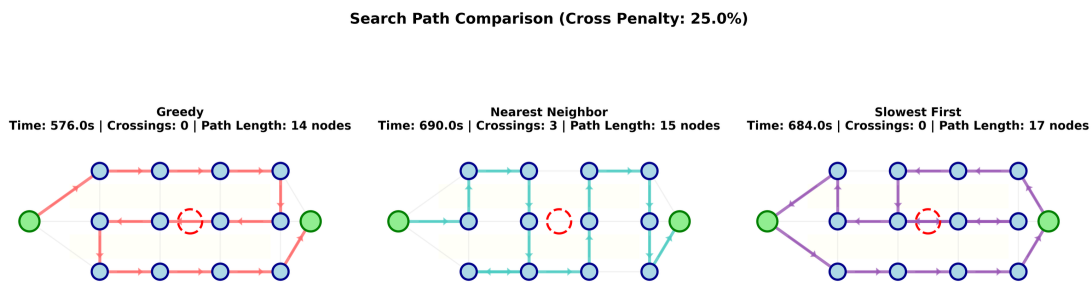


Fig. 13: Graph representation for 12 rooms

In this case, the brute-force algorithm is no longer efficient. Since 12 rooms mean 12! combinations which takes a long time, we dropped it out. Greedy algorithm [5] takes the least time to finish the sweep which seems to mean the local optimization equals to the global optimization. Slowest first aims to sweep the most time-consuming room first avoiding the effect of spread of fire and smoke in the later sweep so that the time on the journey could be reduced.

Secondly, we would extend the existing model to two floors by connecting two exit of one-floor arrangements with certain side weights. The ground floor has 8 rooms and the second floor has 12 rooms reference to the **Fig. 14**.

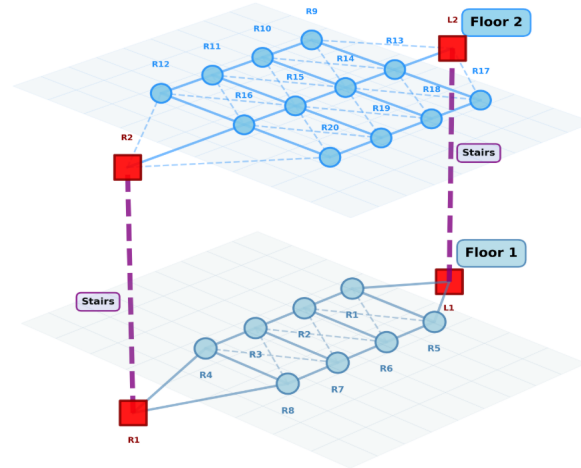


Fig. 14: 3D graph representation of a two-floor layout

Applying the new scenario to the three previously established models, the corresponding routes and time consumption are shown above. The **greedy algorithm** still performs the best, saving a substantial amount of time compared with the other methods. The performance of the **longest-first strategy**, however, appears to be largely theoretical and overly idealized in the previous part. We believe the main reason is that responders spend excessive time locating the rooms ranked highest in the priority queue. Moreover, smoke tends to spread much faster in the upward direction—approximately four to five times faster than on the same floor. As a result, completing the longest tasks first may allow smoke to fill the entire building, which in turn significantly prolongs the subsequent sweep operations.

Two-Floor Building: Search Path Comparison

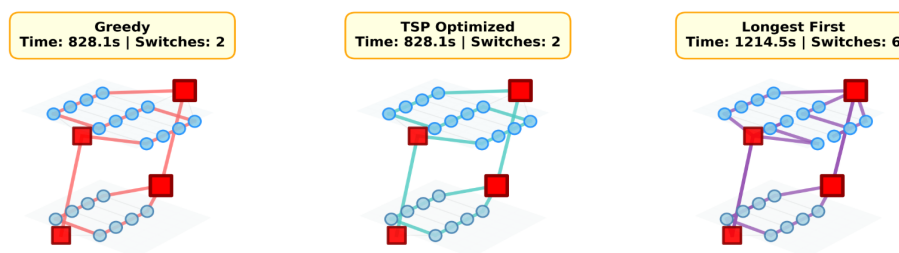


Fig. 15: Responder route in complex scenario under strategies

3.2.4 Team Optimization

Having discussed the influence of increasing the number of teams, we now complement that intuition with a systematic comparison aimed at identifying the optimal team size. We evaluate three searching strategies—Greedy, Longest-First, and TSP. For the TSP strategy, the building is first partitioned into zones, with each responder team assigned to one zone and then following a TSP-optimized route within that area. To capture the slowdown caused when many responders move through the same hallway, we introduce a congestion efficiency, defined as:

$$E_{final} = E_{base} \cdot \exp(-C_{crowding} \cdot (N_{responder} - 1))$$

$$V_{final} = \frac{V_{base}}{E_{final}}$$

where E stands for the congestion efficiency of moving with the value of 1.0 for single team, $C_{crowding}$ represents the crowding factor and $N_{responder}$ is the number of responders in the same hallway.

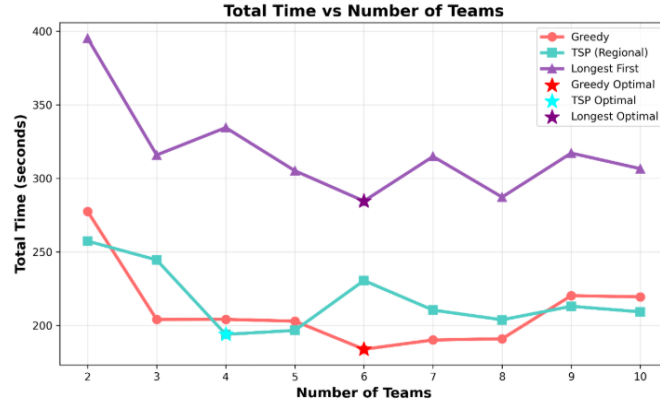


Fig. 16: Variation in Sweeping Time with Increasing Team Numbers

As shown in **Fig. 16**, although the total sweeping time decreases overall, it eventually stabilizes and fluctuates around a relatively consistent value as the number of teams increases. We believe this occurs because adding more teams leads to significant congestion near the building's exits during the initial deployment phase. This bottleneck limits the potential efficiency gains that would otherwise arise from having more teams sweeping rooms concurrently. Based on the comparison, the optimal team size is six for both the Greedy and longest-first strategies, while the TSP-based strategy performs best with four teams.

3.2.5 Reduced Team Approach

From the previous part, the optimal team size varies across different strategies. This suggests that different strategies may be preferable when the number of responders is below the optimal level. Specifically, the TSP strategy performs better with 2, 4, or 5 teams, while the Greedy strategy is optimal for 3 teams or for more than 5 teams, as summarized in Table 3. A single-team scenario is excluded since all strategies yield the same sweeping time.

Table 3: Best Strategies for Limited Number of Teams

Number of Teams	2	3	4	5	≥ 6
Strategies Applied	TSP	Greedy	TSP	TSP	Greedy

4 Data Analysis and Model Extension

4.1 Multi-Source Data Collection and Feature Analysis

Accurate fire risk assessment relies on the systematic analysis and calibration of raw data. The reliability of the model depends on properly processing key elements such as fire temperature, smoke particulates, and toxic gas concentrations to ensure that simulations and predictions reflect realistic building conditions.

High-quality data are essential, sourced from authoritative databases such as NIST fire experiments and NFPA statistics. Key parameters include fire growth coefficients (α), smoke production characteristics and toxic gas yields. Fire growth varies with material and is classified into slow, medium, fast and ultra-fast types. Smoke and fire data focus on temperature and particulate concentration, as these directly affect visibility, safety, and the progression of hazards in the building. Toxic gas data, particularly CO generation rates, are assessed against LC50 and IDLH thresholds to inform safety limits.

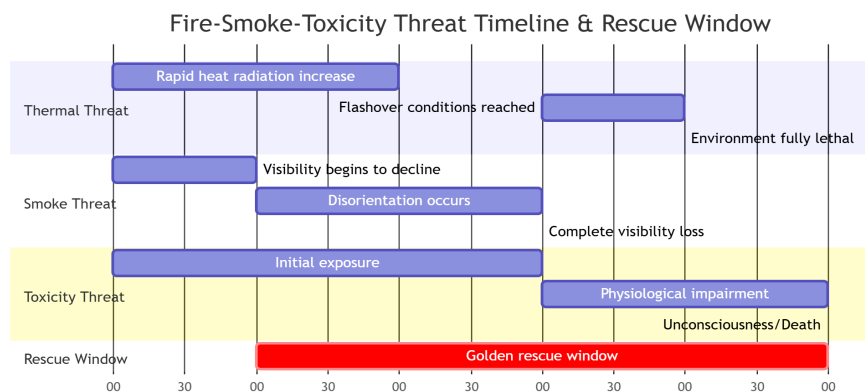


Fig. 17: Fire-Specific Threat Progression and Golden Rescue Window

4.2 Agent-Based Simulation Model

To simulate a more detailed and realistic fire response, we developed an agent-based model (ABM) grounded in the existing numerical parameters. This model integrates the interactions between fire growth and smoke propagation to recreate various fire scenarios, including a simple office, a multi-room office, and a two-floor office layout. Figures illustrate responder behavior under these different conditions.

In the simulation, **blue dots** represent responders, with their remaining health indicated by the color intensity. **Green dots** denote occupants awaiting rescue. Fire sources are represented by **yellow and red dots**, with red fires posing a greater hazard; responders lose health when passing through them. Over time, yellow fires may intensify and become red, increasing the danger. Smoke is visualized in **grey**, spreading outward in a circular pattern from the fire source. The **bright red blocks** on either side indicate exits. Cleared rooms are marked and visually updated. Responders and occupants can endure fire and smoke only briefly; after this limit, they become incapacitated and turn red.

4.2.1 One-Floor Simulation

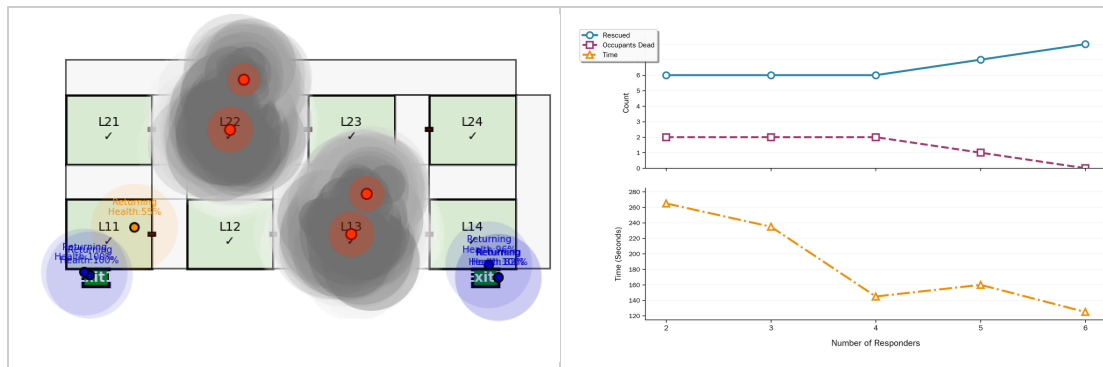


Fig. 18: Simulation Results for Varying Number of Responders

We simulate a single floor with eight rooms and two side exits, with two rooms initially on fire with same structure as shown in 3.2.3. Increasing responders from 2 to 4 reduces sweep time but cannot reach the two trapped occupants. With 5 responders, one occupant can be saved, and with 6, both can be rescued while minimizing risk to responders. This demonstrates that more responders improve sweep efficiency when congestion is not an issue.

4.2.2 Two-Floor Simulation

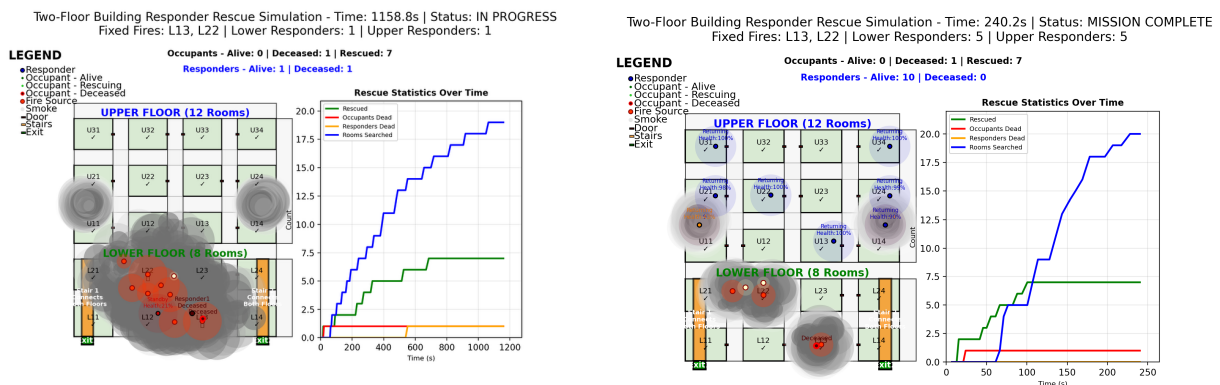


Fig. 19: Comparison of Responder Team Configurations

Moving to the two floor building, the strategies of allocating teams of responders and the total number of the responders was crucial in this scenario.

First, the number of responders varies across simulations. As shown in **Fig 19**, increasing the number of responder teams to nearly five times the original size reduces the task completion time by nearly two-thirds. Overall completion times range widely—from the fastest rescue at 240.2 seconds to scenarios exceeding 1158.8 seconds. Occupant outcomes also differ significantly: Both simulations result in one casualty and seven occupants rescued, with consistent outcomes across different configurations. The responders' operational status likewise varies, with some maintaining full health throughout the mission and others experiencing notable health depletion. These results also indicate that insufficient responder availability can severely worsen outcomes, increasing both the likelihood of fatalities and the

cascading risks posed by fire and smoke.

Next, we examined different allocations of responder teams between the upper and lower floors. This factor is crucial because smoke spreads more rapidly than flames, and real-case evidence shows that most fire-related deaths are caused by suffocation rather than burns. To evaluate the impact of team distribution, multiple simulations were conducted, and two extreme allocations were selected for comparison: an even 4–4 split between floors, and an unbalanced 8–0 allocation that deploys all responders to the lower floor, as shown in the **Fig 20**. Each allocation presents distinct trade-offs. The 8–0 configuration results in one occupant fatality but only one responder sustaining injuries, retaining around 60% of their health. In contrast, the 4–4 allocation avoids occupant deaths and completes the sweep faster, though two responders are injured, with one retaining only 27% health. These results indicate that uneven responder distribution may increase operational risk, especially in unfamiliar buildings with limited visibility and variable smoke conditions.

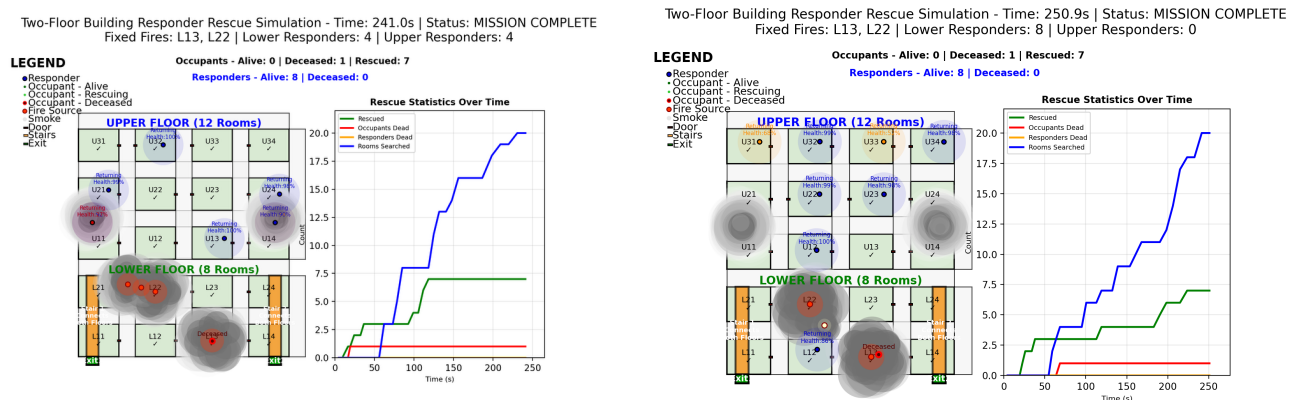


Fig. 20: Comparison of Responder Allocation Strategies

The simulations reveal that both the total number of responders and their allocation strategy are critical to mission success. While increasing personnel improves overall coverage, an effective distribution of teams across floors plays an even more decisive role. In fact, allocation strategy has a greater impact than simply adding more responders: well-configured deployments can improve operational efficiency by more than 30% and reduce casualty rates by nearly half compared to single-floor concentration strategies. These findings highlight that strategic allocation—not just team size—is essential for maximizing rescue effectiveness and minimizing risk.

5 Sensitivity Analysis

5.1 Localized Responder Sweep Model Analysis

To test the robustness of the model, we conducted multiple simulations to calculate the mean number of occupants rescued at each moment, enabling analysis of sweep progress over time.

Additional simulations were performed to evaluate responder cooperation using two complementary strategies: Farthest First and Nearest First, as illustrated in **Fig. 21**.

In the large-scale simulations, the responder using the **Farthest First** strategy (FF1) exhibits more stable and effective performance, prioritizing deep exploration to quickly locate hidden occupants. The **Nearest First** strategy complements this by systematically clearing areas near the entry point. As shown in **Fig. 21**, both curves plateau after approximately 30 seconds, indicating that most single-room operations are completed within this timeframe. Slight fluctuations beyond 30 seconds reflect the careful checking of room corners, primarily contributed by the Nearest First strategy. These results demonstrate that combining complementary search strategies enhances both the efficiency and thoroughness of the sweep.

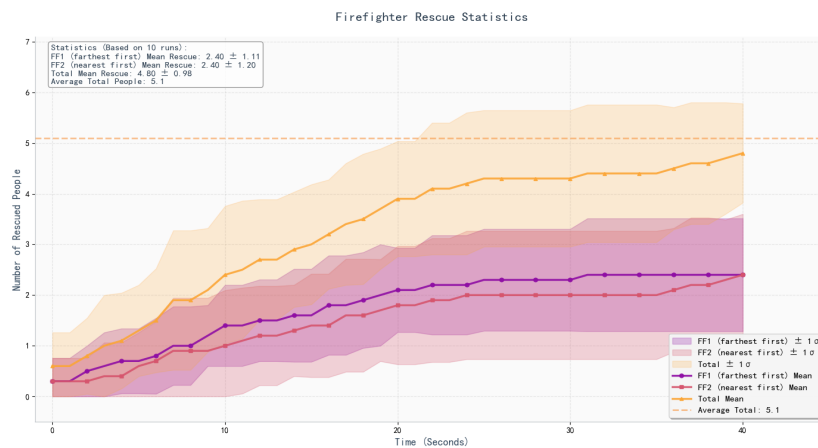


Fig. 21: Comparison of Two Strategies with Error Bars

5.2 Agent-Based Model Analysis

This simulation model involves numerous parameters, as it requires a large amount of input data. To begin the sensitivity analysis, we first identify the most influential parameters and systematically adjust their values to evaluate the robustness of the model and the impact of these changes on the outcomes. The following table shows the core parameters.

Table 5: Core Parameters

Symbol	Parameter name		Tested Range
FF_SPEED	Firefighter	Movement Speed	1.5-2.5
FF_VISION_RADIUS	Firefighter	Vision Radius	2.0-4.0
NEW_FIRE_PROB	New Fire Probability		0.04-0.08
FIRE_SPREAD_RATE	Fire Spread Rate		0.02-0.06
SMOKE_GENERATION_INTERV	Smoke	Generation Interval	0.1-0.5
AL			
MAXIMUM_SMOKE_AGE	Maximum smoke duration		20-80

SMOKE_LETHAL_CONCENTRATION Smoke Lethal 0.5-0.9
TION Concentration

Sensitivity analysis indicates that the model's outcomes are most influenced by the probability of new fire occurrences and the fire spread rate. A higher probability of new fires reduces overall rescue success, as additional fire sources generate more smoke and increase the likelihood of occupant suffocation. Similarly, faster fire spread raises the number of casualties by obstructing responders' paths and decreasing rescue efficiency.

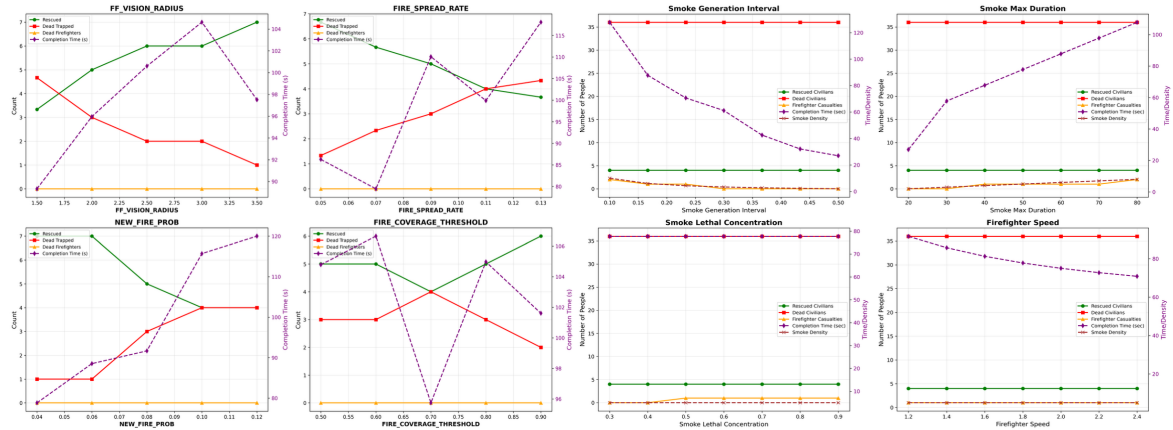


Fig. 22: Fire and Responder Related Parameters and Smoke Related Parameters

The graphs above shows that the model's outcomes are most affected by the smoke generation interval and the maximum smoke duration. Shorter smoke generation intervals significantly prolong rescue times, while longer maximum smoke durations increase the difficulty of clearing areas efficiently.

Despite these sensitivities mentioned above, the model remains robust within realistic operational ranges, enabling effective optimization of responder strategies and resource allocation under varying fire conditions.

6 Conclusion

6.1 Strengths and Weaknesses

Strengths:

1. Our model is comprehensive, incorporating both detailed single-room simulations and overall building-level simulations. This allows the calculated times to serve as reliable references for responders during training exercises.
2. Our model is realistic, accounting for smoke spread, toxic gas, and fire intensity, closely simulating real-world fire scenarios. It can also provide valuable reference for selecting and optimizing firefighting equipment, helping to create a safer operational environment for responders.

Weaknesses:

1. Fire suppression is not explicitly included in the current model. While controlling fire could influence intensity and smoke spread, it adds significant complexity and depends heavily on specific scenarios. For simplicity, this aspect is omitted in the present study, focusing instead on occupant search and rescue dynamics.
2. Clutter is not considered in the model. Such items could be combustible and potentially affect fire behavior. However, to simplify the model, this aspect is not simulated, with the focus placed primarily on the occupant rescue process.

6.2 New Technologies to Consider

Compared with traditional manual, static, isolated models, technology upgrades models by enabling real-time info capture (with smoke-penetrating thermal imaging), data-driven rapid optimal decisions, automated safe execution, and multi-party collaboration—boosting efficiency and reliability.

Technology recommendations should match office layout, area, and density. For a small 2×3 single-floor office (~100–150 m²), use basic sensors, low-power positioning, simplified algorithms, and portable thermal imaging. For a medium 2×4 single-floor office (~150–220 m²), adopt precision detection with thermal imaging, dual-link transmission, and dynamic path algorithms to support multi-responder coordination. For large dual-floor offices (~320–420 m²), implement cross-floor positioning, thermal imaging in stairwells and rooms, fusion algorithms, and full automation. Across all cases, prioritize high-risk areas, balance cost-efficiency, and maintain redundancy through technology and manual backup.

Reference

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- [2] Kobes, M., et al. (2010). Building Safety and Human Behaviour in Fire: A Literature Review. *Fire Safety Journal*, 45(1), 1-11.
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- [6] Kobes, M., et al. (2010). Building Safety and Human Behaviour in Fire: A Literature Review. *Fire Safety Journal*, 45(1), 1-11.

Letter to the Local Emergency Planning Committee

Dear Local Emergency Planning Committee,

On behalf of our team, we are pleased to present the findings and practical recommendations from our comprehensive mathematical modeling project, conducted in response to the Clearing Of Multi-use Assembly Public/Private spaces (COMAP) initiative. Our integrated model aims to address the core challenge of limited information during emergencies and proposes an actionable framework for immediate deployment.

Core Action Strategies

Room-Level Search: Standard Two-Person Operating Procedure

We confirm that the most effective method for room clearance is to deploy two-person teams employing a fixed tactical coordination: one member moves directly to the innermost part of the room, while the other systematically clears the entrance area. Simulations confirm that this pairing minimizes blind spots and avoids redundant efforts, with both clearance efficiency and speed significantly outperforming single-person or other team configurations.

Building-Level Path Planning: Dynamic Risk Assessment

At the building-wide level, we employ a dynamic path algorithm. This algorithm incorporates real-time risks (based on factors such as fire intensity and smoke density) of corridors and other passageways into calculations. For most standard building layouts, the model demonstrates that a "greedy algorithm" efficiently generates near-optimal evacuation paths, outperforming other complex strategies.

Resource Allocation: Scale and Deployment Optimization

Simulations reveal that the number of response teams is not a case of "the more, the better." Exceeding six teams often leads to efficiency degradation due to corridor congestion. The key lies in balanced deployment across floors: in multi-story buildings, concentrating teams on a single floor poses a significant risk. Although balanced allocation may slightly increase the exposure risk for response personnel, it significantly reduces overall casualties, making it the superior choice.

This strategy serves as a data-driven foundation for operations, and its success relies on the integration of tactics, path planning, and resource allocation. We recommend regular drills for calibration and stand ready to provide further technical support.

Sincerely,

Team #16985

AI Report

1 Cursor auto (Nov 15, GPT 5.1, 2025 version)

Query: Increase the font size of titles and labels.

Answer: `ax.set_title(f'{strategy_name}\n'`

`f'Time: {time:.1f}s | Crossings: {crossing_count} | '`

`f'Path Length: {len(path)} nodes',`

`fontsize=50, fontweight='bold', pad=15)`

`=> fontsize=60, fontweight='bold', pad=15)`

2 ChatGPT (Nov 16, GPT 5.1, 2025 version)

Query: There are many parameters for this model since the simulation model requires huge amount of data. First, we have to find out the most influential input and adjust the values of the parameters to test the robustness and the impact of the change these is the start of one model's sensitivity analysis.

Help me to refine is paragraph and fix the gramma mistakes.

Answer: Here's a refined version suitable for an academic context: This simulation model involves numerous parameters, as it requires a large amount of input data. To begin the sensitivity analysis, we first identify the most influential parameters and systematically adjust their values to evaluate the robustness of the model and the impact of these changes on the outcomes.